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RA-11: Linear-accuracy trace estimation for arbitrary PSD inputs

Hierarchical partial-trace control, finite randomness,
and fixed-order optimality

Prepared with ChatGPT; proofs, exact identity checks, and reproducible diagnostics

Status: a stronger partial resolution, not a full solution of RA-11. For every fixed tensor order q , the new argument matches the minimax dependence on n and ε for *all* PSD inputs. The constants still grow exponentially with q , and the unrestricted lower bound does not match that growth. RA-11 asks for universal constant factors simultaneously in all three parameters, so that final requirement remains unresolved here. The manuscript has not received independent review or formal proof-assistant verification.

Abstract

Let $M \succeq 0$ be an arbitrary real matrix of order $N = n^q$, accessible only through real rank-one tensor matrix-vector products. We construct a finite-randomness adaptive trace estimator with relative error ε and success probability at least $5/6$ using at most

$$\min \left\{ n^q, \left\lceil \frac{2^{71} q(q+1)(2^q-1)}{\varepsilon} \left\lceil \sqrt{(3-1/n)^q - 1} \right\rceil \right\rceil \right\}$$

original oracle calls. In particular, the dependence on accuracy is linear in $1/\varepsilon$, with no dimension-dependent multiplier for fixed q . The construction learns low-rank controls for the original PSD partial traces, using residual contractions as unbiased noisy matrix-vector products. An orthogonal interaction decomposition identifies which approximation error controls each residual component and bounds all later contraction noise by the accumulated squared errors. Two-sided sketching, deterministic column-subset selection, and metric-median boosting supply the required noisy approximation guarantee. Internal four-phase complex probes are simulated by $q+1$ real product calls through polynomial interpolation. Together with an ordinary-oracle lower bound, this yields $Q(n, q, \varepsilon) = \Theta_q(\min\{n^q, \varepsilon^{-1}\})$. The remaining dependence on q is separated explicitly from this fixed-order result.

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1 Target, progress, and remaining gap

1.1 The exact model

Fix integers $n, q \geq 2$, let $N = n^q$, and take $0 < \varepsilon < 1/2$. An unknown real symmetric matrix $M \in \mathbb{R}^{N \times N}$ is PSD. A query specifies $v_1, \dots, v_q \in \mathbb{R}^n$ and returns the *entire* vector

$$M(v_1 \otimes \dots \otimes v_q).$$

Let $Q(n, q, \varepsilon)$ be the smallest worst-case number of calls for a randomized adaptive algorithm satisfying, for every fixed $M \succeq 0$,

$$\mathbb{P}\{|\hat{\tau} - \text{tr } M| \leq \varepsilon \text{tr } M\} \geq 2/3. \quad (1)$$

The model permits arbitrary finite exact arithmetic and imposes no query-conditioning bound. The matrix M itself is *not* assumed to be a tensor product. These are the requirements in RA-11 [1].

The scalar conditioned-oracle lower bound in Meyer–Swartworth–Woodruff [4] is not used as a lower bound for this full-vector oracle. The partial-trace moment viewpoint of Meyer–Avron [3] and ordinary variance-reduction methods such as Hutch++ [5] are relevant precedents, but do not by themselves implement the algorithm below in the permitted oracle.

1.2 Main new theorem

Define

$$K = 2^q - 1, \quad h_s = (2 - 1/n)^s, \quad H = (3 - 1/n)^q - 1, \quad D = \lceil \sqrt{H} \rceil. \quad (2)$$

In particular, $\sum_{\emptyset \neq S \subseteq [q]} h_{|S|} = H$.

Theorem 1.1 (Linear accuracy dependence for arbitrary PSD inputs). *In the real full-vector product-query model,*

$$Q(n, q, \varepsilon) \leq \min \left\{ n^q, \left\lceil \frac{2^{71} q(q+1) K D}{\varepsilon} \right\rceil \right\}. \quad (3)$$

The constructed estimator has success probability at least 5/6. All its randomness can be generated by finitely many fair bits. Every actual oracle input is a real product vector; no call is made with an output-derived unstructured vector.

The constant 2^{71} is intentionally conservative. It is a checked sufficient constant, not an optimized value or a statement of practical efficiency. Since $D \leq 2\sqrt{H}$ here, the new branch is

$$Q(n, q, \varepsilon) = O \left(\min \left\{ n^q, \frac{q(q+1)[2\sqrt{3-1/n}]^q}{\varepsilon} \right\} \right), \quad (4)$$

with a universal implied constant.

Corollary 1.2 (Fixed tensor order). *For each fixed $q \geq 2$, uniformly over $n \geq 2$ and $0 < \varepsilon < 1/2$,*

$$Q(n, q, \varepsilon) = \Theta_q \left(\min \{ n^q, \varepsilon^{-1} \} \right). \quad (5)$$

The constants in Θ_q are allowed to depend on q ; this qualification is essential.

Proof. The upper bound follows from Theorem 1.1, since K, D are bounded by functions of q alone. Appendix A supplies the universal lower bound $Q \geq \min(N, 1/\varepsilon)/80000$ even when arbitrary ordinary matrix-vector queries are allowed. That stronger oracle simulates every permitted product query. $\square$

For example, with $q = 2$ and $\varepsilon = n^{-1/2}$ as $n \rightarrow \infty$, this gives $Q = \Theta(\sqrt{n})$. The previous package's general upper bounds only supplied $O(n)$ in that regime. Unlike the earlier tensor-product-matrix theorem, the new statement applies to arbitrary PSD inputs.

1.3 Combined envelope and what is not proved

The first package [2] established

$$L(n, q, \varepsilon) = \max_{1 \leq b \leq q} \min \left\{ n^b, \frac{\sqrt{\lfloor q/b \rfloor}}{\varepsilon} \right\}, \quad Q \geq L/80000.$$

Its four upper branches were

$$U_0 = n^q, \quad U_{\mathbb{R}} = \left\lceil 3((3n/(n+2))^q - 1)/\varepsilon^2 \right\rceil, \quad (6)$$

$$U_{\mathbb{C} \rightarrow \mathbb{R}} = (q+1) \left\lceil 3((2n/(n+1))^q - 1)/\varepsilon^2 \right\rceil, \quad U_{\text{ord}} = n^{q-1}(5 \lceil 4/\varepsilon \rceil + 4). \quad (7)$$

The updated valid envelope is therefore

$$\frac{L}{80000} \leq Q \leq \min \left\{ U_0, U_{\mathbb{R}}, U_{\mathbb{C} \rightarrow \mathbb{R}}, U_{\text{ord}}, \left\lceil 2^{71} q(q+1) K D / \varepsilon \right\rceil \right\}. \quad (8)$$

The full blocking lower bound and the separate product-matrix characterization are preserved in the included first-round archive; they are not claimed as new here.

The unresolved requirement is the joint dependence on q . For instance, at $n = 2$, $\varepsilon = 1/4$, the retained bounds give a lower scale $\Omega(\sqrt{q/\log q})$ and an upper scale $O(q(4/3)^q)$ for large q . The new linear-accuracy theorem does not close that gap. Consequently neither (3) nor (5) is a full universal-constant solution of RA-11.

2 Four-phase probes and their real implementation

2.1 Finite random bits

A local four-phase vector $z_i \in \mathbb{C}^n$ has independent entries uniform in $\{1, -1, i, -i\}$. Products of these vectors will be called phase-product probes. Complex numbers are stored as pairs of reals.

Lemma 2.1 (Isotropy, properness, and fourth moments). *If $z = z_1 \otimes \cdots \otimes z_s$, then*

$$\mathbb{E} z z^* = I, \quad \mathbb{E} z z^T = 0, \quad \mathbb{E} |u^* z|^4 \leq h_s \|u\|_2^4 \quad (u \in \mathbb{C}^{n^s}), \quad h_s = (2 - 1/n)^s.$$

Also $\|z_i\|_2^2 = n$ deterministically.

Proof. For one factor, independence and the first two phase moments give isotropy and properness. Direct expansion gives

$$\mathbb{E} |u^* z_i|^4 = 2 \|u\|_2^4 - \sum_j |u_j|^4 \leq (2 - 1/n) \|u\|_2^4.$$

For the induction, contract all but one factor and call the resulting coefficient vector w . Conditional on w , the preceding inequality bounds the fourth moment by $(2 - 1/n) \|w\|_2^4$. If u_j are the slices of u in that factor, then Cauchy–Schwarz and the induction hypothesis imply

$$\mathbb{E}|w_j|^2 |w_k|^2 \leq h_{s-1} \|u_j\|_2^2 \|u_k\|_2^2.$$

Summing j, k proves the bound. Isotropy and properness tensorize directly. $\square$

2.2 No complex oracle is assumed

Lemma 2.2 (Real interpolation simulation). *For any real linear map M and any complex factors $z_i = a_i + ib_i$, $M(z_1 \otimes \cdots \otimes z_q)$ is computable using $q + 1$ real product queries.*

Proof. The vector polynomial $p(t) = \bigotimes_{i=1}^q (a_i + tb_i)$ has degree at most q . At the distinct real nodes $0, \dots, q$, form $Mp(j)$ and interpolate:

$$Mp(i) = \sum_{j=0}^q \left(\prod_{\substack{0 \leq h \leq q \\ h \neq j}} \frac{i - h}{j - h} \right) Mp(j). \quad (9)$$

The coefficients are Gaussian rational. All actual inputs have real factors, and all computation is finite. The identity is valid for adaptively chosen complex factors. $\square$

This reduction was already proved in the first package [2]. It is recalled because its $q + 1$ cost must be charged in the new theorem. Interpolation may be poorly conditioned in floating-point arithmetic; exact-real query complexity is the model here.

3 Partial traces and orthogonal interactions

3.1 Definitions and deterministic update geometry

For $S \subseteq [q]$, write $\mathcal{T}_S R = \text{tr}_{S^c} R$ for the unnormalized partial trace retaining modes S , in increasing order. Set

$$\mathcal{L}_S A = A \otimes I_{S^c} / n^{q-|S|}, \quad \mathcal{E}_S = \mathcal{L}_S \mathcal{T}_S,$$

with the factors permuted back to their original positions. For $S = \emptyset$, $\mathcal{E}_\emptyset R = (\text{tr } R)I/N$.

Lemma 3.1 (Commuting conditional expectations). *In Frobenius inner product $\langle A, B \rangle = \text{tr}(A^* B)$, $\mathcal{E}_S$ is an orthogonal projection, and*

$$\mathcal{E}_S \mathcal{E}_T = \mathcal{E}_{S \cap T}, \quad \mathcal{T}_T \mathcal{E}_S = \mathcal{E}_{S \cap T}^{(T)} \mathcal{T}_T. \quad (10)$$

Here the superscript indicates an operator on the retained T -space. Moreover,

$$\|\mathcal{T}_T \mathcal{L}_S e\|_F^2 \leq n^{|S|-|T|} \|e\|_F^2. \quad (11)$$

Proof. On each local matrix factor, the map is either the identity or the orthogonal projection $A \mapsto (\text{tr } A)I/n$. Tensoring these commuting projections proves (10). For (11), tracing the $S \setminus T$ modes has Frobenius operator norm $n^{|S \setminus T|/2}$, and lifting through $T \setminus S$ has norm $n^{-|T \setminus S|/2}$. The squared product is $n^{|S|-|T|}$. $\square$

Maintain a known Hermitian control B , and put $R = M - B$. If $\hat{A}_S$ estimates $A_S = \mathcal{T}_S M$, update

$$B' = B + \mathcal{L}_S(\hat{A}_S - \mathcal{T}_S B). \quad (12)$$

Writing $e_S = A_S - \hat{A}_S$ gives

$$R' = (I - \mathcal{E}_S)R + \mathcal{L}_S e_S, \quad \mathcal{T}_S R' = e_S. \quad (13)$$

The two terms after applying $\mathcal{T}_T$ are orthogonal, by (10). Consequently, whenever $|T| \geq |S|$,

$$\|\mathcal{T}_T R'\|_F^2 \leq \|\mathcal{T}_T R\|_F^2 + \|e_S\|_F^2. \quad (14)$$

This already gives a useful descending-order error invariant. The next decomposition provides a sharper variance budget than simply summing (14) over T .

3.2 Interaction components

On each local matrix space, split orthogonally into scalar multiples of I and traceless matrices. For $V \subseteq [q]$, let $\mathcal{P}_V$ project onto tensors whose traceless factors occur precisely in V . Equivalently,

$$\mathcal{P}_V = \sum_{T \subseteq V} (-1)^{|V|-|T|} \mathcal{E}_T. \quad (15)$$

These are mutually orthogonal projections summing to the identity, and

$$\mathcal{E}_S = \sum_{V \subseteq S} \mathcal{P}_V. \quad (16)$$

The component $R_V = \mathcal{P}_V R$ can be written $\mathcal{L}_V C_V$, where C_V has zero partial trace in every retained mode. Also $\|R_V\|_F^2 = n^{-(q-|V|)} \|C_V\|_F^2$.

Lemma 3.2 (Which error controls each component). *Process all nonempty sets in any order of nonincreasing cardinality, using (12). After the first, full-set update, fix a component V . Its current value is the V -component of $\mathcal{L}_U e_U$, where U is the most recently processed set containing V . Immediately before processing S , if $V \not\subseteq S$, this set U satisfies*

$$|U| \leq |S \cup V|. \quad (17)$$

After all nonempty sets have been processed, every nonempty component V is controlled by its own error e_V .

Proof. By (16), update S replaces exactly those interaction components indexed by $V \subseteq S$, and leaves all others unchanged. The first update replaces the whole residual by $e_{[q]}$, proving the first assertion by induction. Put $s = |S|$. If $|V| > s$, set V itself has already been processed, and no later set can properly contain V , so $U = V$. If $|V| \leq s$ and $V \not\subseteq S$, a set of size $s+1$ containing V has already been processed (the case $s = q$ cannot occur here). Thus the most recent such U has size at most $s+1$, while $|S \cup V| \geq s+1$. This proves (17). Once V itself is processed, no distinct subsequent set can contain V ; the last assertion follows. $\square$

4 Contraction noise is paid for by earlier errors

For a product phase vector y on S^c , let $K_y x = x \otimes y$, ordered in the original modes, and define the random compressed matrix

$$C_y(R) = K_y^* R K_y.$$

Its expectation is $\mathcal{T}_S R$. Full-vector responses permit computing $K_y^* R K_y x$ by output contractions; the entire compressed matrix is not queried or assumed known.

Lemma 4.1 (One-mode contraction). *For any complex square matrix F and a local phase vector z ,*

$$\mathbb{E} |z^* F z - \text{tr } F|^2 = \sum_{a \neq b} |F_{ab}|^2 \leq \|F\|_F^2.$$

Consequently, contracting one mode of a matrix R gives

$$\mathbb{E} \|C_z(R)\|_F^2 \leq \|\text{tr}_i R\|_F^2 + \|R\|_F^2. \quad (18)$$

These claims do not require R to be PSD.

Proof. Expand using independent phases, $\mathbb{E} z_a = \mathbb{E} z_a^2 = 0$, and $|z_a| = 1$. Only the matched off-diagonal index pairs survive the variance expansion. Apply the scalar identity to every retained matrix entry and sum to obtain (18). $\square$

Lemma 4.2 (Orthogonal interaction noise bound). *For Hermitian $R = \sum_V R_V$,*

$$\mathbb{E} \|C_y(R) - \mathcal{T}_S R\|_F^2 \leq \sum_{V \not\subseteq S} n^{q-|S \cup V|} \|R_V\|_F^2. \quad (19)$$

If R is the residual immediately before a non-full stage S in the descending updates of Lemma 3.2, then

$$\mathbb{E} \|C_y(R) - \mathcal{T}_S R\|_F^2 \leq \sum_{U \text{ already processed}} \|e_U\|_F^2. \quad (20)$$

For the final residual and a full product phase probe z ,

$$\text{Var}(z^* R z) \leq \sum_{\emptyset \neq S \subseteq [q]} \|e_S\|_F^2. \quad (21)$$

Proof. If $V \subseteq S$, every outside factor of R_V is proportional to I/n . Since $\|z_i\|^2 = n$, its contraction is deterministic and equals $\mathcal{T}_S R_V$. If $V \not\subseteq S$, at least one outside factor is traceless, so the contraction has mean zero.

Distinct interaction components have zero cross inner product in L^2 of the Frobenius space after contraction. If their retained supports differ, one has a traceless retained factor where the other has an identity factor, giving pointwise Frobenius orthogonality. Otherwise their outside supports differ, and conditioning on a differing outside mode gives mean zero for one component and a deterministic identity contraction for the other. This proves orthogonality of all the contributing random matrices.

Write $R_V = \mathcal{L}_V C_V$. Contracting the $V \setminus S$ modes of C_V does not increase expected squared Frobenius norm: apply (18) successively, observing that the appropriate partial traces remain zero. Lifting through the retained $S \setminus V$ identity modes contributes $n^{-|S \setminus V|}$. Thus

$$\mathbb{E} \|C_y(R_V)\|_F^2 \leq n^{-|S \setminus V|} \|C_V\|_F^2 = n^{q-|S \cup V|} \|R_V\|_F^2.$$

Summing the orthogonal components proves (19).

For (20), assign each $V \not\subseteq S$ to its current set U from Lemma 3.2. Its weighted squared norm in (19) is

$$n^{|U|-|S \cup V|} \left\| \mathcal{P}_V^{(U)} e_U \right\|_F^2 \leq \left\| \mathcal{P}_V^{(U)} e_U \right\|_F^2.$$

For each fixed U , the projections on the right are mutually orthogonal. Summing those assigned to U gives at most $\|e_U\|_F^2$, proving the prefix bound. At the end, take $S = \emptyset$. The scalar component is deterministic; every nonempty V is assigned to $U = V$, so its factor is one. This proves (21). $\square$

4.1 The noisy oracle has the correct PSD mean

At stage S , freeze the known control B and let $A_S = \mathcal{T}_S M$. For an inside product vector x , use a independent outside probes and return

$$\frac{1}{a} \sum_{j=1}^a K_{y_j}^* (M - B) K_{y_j} x + (\mathcal{T}_S B) x. \quad (22)$$

This equals $(A_S + \overline{F})x$, where $\mathbb{E} \overline{F} = 0$, $\overline{F}$ is Hermitian, independent of x , and

$$\mathbb{E} \left\| \overline{F} \right\|_F^2 \leq \frac{1}{a} \sum_{U \text{ processed}} \|e_U\|_F^2. \quad (23)$$

The matrices $\overline{F}$ in different calls are independent copies conditional on the frozen history. Each outside sample uses one complex full product call and hence $q + 1$ original real calls. Applying or partially tracing B is computation on a known matrix and costs no oracle calls.

The mean is the original PSD partial trace A_S , not $\mathcal{T}_S(M - B)$. Therefore $A_S \succeq 0$ and $\text{tr } A_S = \tau$ at every stage even though the residual and the controls can be indefinite. This distinction is needed in the approximation lemma that follows.

5 Noisy PSD approximation from permitted sketches

5.1 A lower covariance estimate

Lemma 5.1 (Elementary covariance injection). *Suppose $g \in \mathbb{C}^r$ satisfies $\mathbb{E} g g^* = I$, $\mathbb{E} g g^T = 0$, and $\mathbb{E} |u^* g|^4 \leq h \|u\|^4$. For $m \geq 2^{25} h r$ independent copies,*

$$\mathbb{P} \left\{ \frac{1}{m} \sum_{j=1}^m g_j g_j^* \succeq \frac{1}{4} I \right\} \geq 63/64.$$

Proof. First consider an isotropic real vector $w \in \mathbb{R}^r$ with fourth-moment bound h . For a unit u , put $\phi(u^T w) = \min\{(u^T w)^2, 4h\}$. Then $\mathbb{E} \phi(u^T w) \geq 3/4$, since

$$\mathbb{E}[(u^T w)^2 \mathbf{1}_{(u^T w)^2 > 4h}] \leq \frac{\mathbb{E}(u^T w)^4}{4h} \leq 1/4.$$

The scalar function ϕ is $4\sqrt{h}$ -Lipschitz and vanishes at zero. Symmetrization and the scalar Rademacher contraction inequality give

$$\mathbb{E} \sup_{\|u\|=1} \left| \frac{1}{m} \sum_j \phi(u^T w_j) - \mathbb{E} \phi(u^T w) \right| \leq 16\sqrt{h} \mathbb{E} \left\| \frac{1}{m} \sum_j \epsilon_j w_j \right\|_2$$

$$\leq 16\sqrt{hr/m}.$$

The first step uses symmetrization factor two and contraction factor two for the absolute supremum; the second uses independence of the Rademacher signs and $\mathbb{E}\|w\|^2 = r$. These inequalities apply to finite nets and then, by continuity and domination by a bounded function, to the unit sphere. For $m \geq 2^{22}hr$, Markov's inequality makes the displayed supremum at most $1/2$ except with probability $1/64$. Hence every empirical clipped second moment is at least $1/4$, and the unclipped second moment is larger.

For the complex claim, realify g as $w = \sqrt{2}(\Re g, \Im g) \in \mathbb{R}^{2r}$. Properness makes this vector isotropic, and its real fourth-moment bound is at most $4h$. The real result needs $2^{22}(4h)(2r) = 2^{25}hr$ samples. Apply its Gram inequality to the real and imaginary linear forms of each u^*g and add; the complex Gram bound follows. $\square$

Fourth-moment lower-covariance bounds are established tools; see, for example, Oliveira [6]. The deliberately loose constant above is derived here so that no unquantified subspace-embedding assertion is needed.

5.2 Column selection without exact singular vectors

Lemma 5.2 (Finite column selection). *For any complex matrix Y and integer $r_0 \geq 1$, one can select at most $2r_0$ columns whose orthogonal projector P satisfies*

$$\|(I - P)Y\|_F^2 \leq 2\|Y - Y_{r_0}\|_F^2. \quad (24)$$

Selection can be done by finite arithmetic, exact rank tests, and exhaustive comparison of projection residuals; no new oracle calls or exact SVD are required.

Proof. If $\text{rank } Y \leq 2r_0$, select a basis of its columns. Otherwise put $k = 2r_0$. For every independent k -column submatrix Y_J , assign weight $w_J = \det(Y_J^* Y_J)$ and let P_J project onto its span. A Gram determinant Schur-complement identity gives

$$\sum_{|J|=k} w_J \|(I - P_J)Y\|_F^2 = (k+1)e_{k+1}, \quad \sum_{|J|=k} w_J = e_k,$$

where e_j is the j th elementary symmetric polynomial of the squared singular values $\lambda_1 \geq \lambda_2 \geq \dots$ of Y . Indeed each $(k+1)$ -column determinant is counted once for each of its $k+1$ omitted columns in the first sum. Cauchy–Binet identifies the sums of principal minors with e_j .

Every $(k+1)$ -element index set contains at least $k - r_0 + 1$ indices larger than r_0 . Counting their contributions to $e_k \sum_{j>r_0} \lambda_j$ gives

$$(k - r_0 + 1)e_{k+1} \leq e_k \sum_{j>r_0} \lambda_j.$$

Therefore the weighted mean projection error is at most $(k+1)/(k - r_0 + 1)$ times the best rank- r_0 squared residual, and this ratio is less than two. At least one subset is no worse than its weighted mean. Exhaustive minimization over the finitely many independent subsets finds one. Every residual is computable from $Y_J(Y_J^* Y_J)^{-1} Y_J^*$. $\square$

The column-count/error tradeoff is a standard column-subset result [7]; the argument above also supplies its complex version and the finite-arithmetic implementation used here. The search may be prohibitively expensive, but arithmetic cost is not charged in RA-11.

5.3 Two-sided noisy reconstruction

Lemma 5.3 (Noisy PSD approximation). *Let $A \succeq 0$ be Hermitian of order d , with trace τ . Suppose inputs can be drawn from an isotropic proper complex distribution with fourth bound h , and a noisy call on x returns $(A + F)x$, where F is an independent copy of a fixed mean-zero Hermitian random matrix, independent of x , with $\mathbb{E} \|F\|_F^2 \leq \nu$. For each integer $r_0 \geq 1$, a finite-arithmetic algorithm using at most $2^{28}hr_0$ noisy calls returns a Hermitian $\hat{A}$ such that*

$$\mathbb{P} \left\{ \left\| \hat{A} - A \right\|_F^2 \leq 2^{13}(\tau^2/r_0 + \nu) \right\} \geq 7/8. \quad (25)$$

A coordinate basis may be used instead of random inputs when $r_0 \geq d$.

Proof. If $r_0 \geq d$, query a basis and Hermitize the resulting noisy matrix. Identically distributed noise matrices give expected squared error $\sum_j \mathbb{E} \|F_j e_j\|^2 = \mathbb{E} \|F\|_F^2 \leq \nu$; Markov proves the stated bound.

Now suppose $r_0 < d$. Put $J_0 = 2^{25}$ and $m = \lceil J_0 h r_0 \rceil$. For analysis, normalize all sketch columns by $1/\sqrt{m}$ and write

$$Y = A\Omega + E.$$

Select the at most $2r_0$ columns provided by Lemma 5.2, and let P be their orthogonal projector. An orthonormal basis U for its range is used only in the proof. Put $k = \text{rank } P \leq 2r_0$.

Diagonalize A for analysis as $A = U_1 \Lambda_1 U_1^* + U_2 \Lambda_2 U_2^*$, splitting after r_0 eigenvalues. Set $A_2 = U_2 \Lambda_2$ and $\Omega_j = U_j^* \Omega$, and put $t_0 = \|A - A_{r_0}\|_F^2 \leq \tau^2/r_0$. With probability at least $63/64$, the top block Ω_1 has $\|\Omega_1^\dagger\|_2 \leq 2$, by Lemma 5.1. Also

$$\mathbb{E} \|A_2 \Omega_2\|_F^2 = t_0, \quad \mathbb{E} \|E\|_F^2 \leq \nu.$$

Thus each is at most 64 times its mean bound except on an event of probability $1/64$ (a zero mean bound gives zero almost surely).

The column selection guarantee implies

$$\|(I - P)Y\|_F^2 \leq 2 \|Y - Y_{r_0}\|_F^2 \leq 4(\|A_2 \Omega_2\|_F^2 + \|E\|_F^2).$$

Reconstructing the leading block through $\Omega_1^\dagger$, and bounding each sum of two matrices by twice the sum of squared norms, gives

$$\begin{aligned} \|(I - P)A\|_F^2 &\leq t_0 + 48(\|A_2 \Omega_2\|_F^2 + \|E\|_F^2) \\ &\leq 3073t_0 + 3072\nu. \end{aligned} \quad (26)$$

For clarity, the first factor 48 comes from $8\|(I - P)Y\|_F^2 + 8\|A_2 \Omega_2 + E\|_F^2$ for the leading-block error, followed by the preceding display. No independence between spectral blocks of Ω has been assumed.

If $k = 0$, return zero; (26) already suffices on these events. Otherwise take an independent normalized sketch Ψ with $m' = \lceil J_0 h k \rceil$ columns, observe $Z = A\Psi + F'$, and put

$$G = U^* \Psi, \quad \tilde{A} = U(GG^*)^{-1} G Z^*.$$

If the Gram matrix is singular, return zero on that branch. Conditional on all first-stage data, it is nonsingular and has inverse norm at most four except with probability $1/64$. Let $R_0 = (I - P)A$. Since $U^*R_0 = 0$, independent columns and fourth-moment Cauchy–Schwarz give

$$\mathbb{E} \|U^*\Psi\Psi^*R_0\|_F^2 \leq \frac{hk}{m'} \|R_0\|_F^2. \quad (27)$$

Here each entry is an average of centered products; for vectors u, v , $\mathbb{E}|u^*x|^2|v^*x|^2 \leq h\|u\|^2\|v\|^2$. Also $\mathbb{E}\|F'\|_F^2 \leq \nu$. Markov again bounds both squared norms by 64 times their respective right sides. The residual orthogonal to U and the fitted coefficient error are Frobenius orthogonal, so

$$\begin{aligned} \|\tilde{A} - A\|_F^2 &\leq \left(1 + \frac{2048hk}{m'}\right) \|R_0\|_F^2 + 512\nu \\ &\leq 6146t_0 + 6656\nu \leq 2^{13}(\tau^2/r_0 + \nu). \end{aligned}$$

The noise term uses $\|(GG^*)^{-1}G\|_2^2 \leq 4$; it does not require independence between the noise columns and their input vectors. Hermitization $(\tilde{A} + \tilde{A}^*)/2$ cannot increase error relative to Hermitian A .

The six exceptional events total at most $6/64 < 1/8$. Finally, $m \leq 2J_0hr_0$, $m' \leq 4J_0hr_0$, and $m + m' \leq 6J_0hr_0 < 2^{28}hr_0$. $\square$

Remark 5.4 (Actual arithmetic and actual inputs). An arbitrary independent-column basis F_0 of the selected columns can replace U . The exact formula is

$$\tilde{A} = F_0(F_0^*\Psi\Psi^*F_0)^{-1}F_0^*\Psi Z^*.$$

Using unnormalized raw sketches gives the same result, since normalization factors cancel. Rank tests, inverses, projections, and comparisons of squared residuals use only finitely many real arithmetic operations and comparisons. No spectral decomposition or square root is performed by the exact algorithm. The random inputs are product probes, and the basis-sweep inputs are coordinate products. In particular, the algorithm never requests AU or AF_0 .

5.4 Boosting without the unknown trace

Lemma 5.5 (Metric-median boosting). *Run Lemma 5.3 independently $\ell_0 = 8q+17$ times, conditional on the frozen stage history. Among the candidates, select one whose median squared Frobenius distance to the others (including itself) is smallest. Then, except with probability at most $1/(12K)$,*

$$\|\hat{A} - A\|_F^2 \leq 2^{17}(\tau^2/r_0 + \nu). \quad (28)$$

Selection needs neither τ nor a known good-error radius.

Proof. If a strict majority lies in the radius- σ ball around A , a good candidate has median distance at most 2σ . The selected candidate therefore has at least half its neighbors within 2σ , including some good candidate, so it lies within 3σ of A . The squared bound is multiplied by at most nine, and $9 \cdot 2^{13} < 2^{17}$. Each independent failure probability is at most $1/8$. Hoeffding’s inequality gives probability at most $e^{-9\ell_0/32}$ for half or more failures. Since $9\ell_0/32 \geq 3 + q > \log(12K)$ for $q \geq 2$, this is at most $1/(12K)$. $\square$

6 Complete algorithm and proof of the upper bound

We now give the parameters of a finite algorithm. These large values certify the bound; the experimental implementation uses explicitly smaller, uncertified values. The dimension-independent hierarchy is compared in advance with the N -query coordinate-basis algorithm, and the cheaper certified branch may be selected.

6.1 The large-error branch

If $\varepsilon > 1/D$, use $m = \lceil 6K/\varepsilon^2 \rceil$ independent full phase-product probes and average $z^* M z$. Lemma 2.1, a PSD eigenvalue decomposition, and Cauchy–Schwarz imply

$$\mathbb{E}(z^* M z)^2 \leq h_q \tau^2, \quad \text{Var}(z^* M z) \leq (h_q - 1) \tau^2 \leq K \tau^2.$$

Chebyshev gives failure at most $1/6$. The number of complex calls is at most $6KD/\varepsilon + 1 \leq 7KD/\varepsilon$.

6.2 The hierarchy branch

Suppose $\varepsilon \leq 1/D$ and set

$$r = \left\lceil \frac{1}{D\varepsilon} \right\rceil, \quad r_0 = 2^{18}r, \quad a = 2^{18}K, \quad \ell_0 = 8q + 17, \quad \ell = \left\lceil \frac{12K}{r\varepsilon^2} \right\rceil. \quad (29)$$

Initialize $B = 0$. Process every nonempty $S \subseteq [q]$ by nonincreasing size. During a stage, freeze B and perform the following operations:

1. Use (22) to provide unbiased noisy responses for $A_S = \mathcal{T}_S M$. Use a outside samples per response, or one when $S = [q]$.
2. Apply the boosted approximation procedure with rank target r_0 and fourth bound $h_{|S|}$, obtaining $\hat{A}_S$.
3. Update the known control with (12).

After all stages, take ℓ fresh full phase-product probes and return

$$\hat{\tau} = \text{tr } B + \frac{1}{\ell} \sum_{j=1}^{\ell} z_j^* (M - B) z_j. \quad (30)$$

All values in (30) are real because the controls are Hermitian.

Proof of Theorem 1.1. Take a fixed nonzero $M \succeq 0$ and write $\tau = \text{tr } M > 0$. Call a stage successful when

$$\|e_S\|_F^2 \leq \tau^2/r. \quad (31)$$

For the full stage, the noisy oracle is exact. At any later stage, condition on all previous stages being successful and on their complete history. Lemma 4.2 bounds the unaveraged noise second moment by $K\tau^2/r$. Averaging a independent samples and using (29) gives

$$\nu \leq \frac{\tau^2}{2^{18}r}, \quad \frac{\tau^2}{r_0} = \frac{\tau^2}{2^{18}r}.$$

Therefore (28) implies (31), except with conditional probability at most $1/(12K)$. A first-failure union bound over K stages shows that every stage succeeds with probability at least $11/12$; independence across stages is neither assumed nor needed.

Condition on a successful completed hierarchy. Equation (21) bounds the final single-probe residual variance by $K\tau^2/r$. The independent average in (30) has conditional mean τ and variance at most $\varepsilon^2\tau^2/12$ by the choice of ℓ . Chebyshev gives conditional failure at most $1/12$. Total failure is at most $1/12 + 1/12 = 1/6$. The zero input gives identically zero controls and output, so it is handled exactly.

It remains to count queries. At stage S , the boosted approximation uses at most $2^{28}h_{|S|}r_0\ell_0$ noisy responses. Each costs at most a complex product calls, including the full stage where this upper bound is loose. Thus all stages cost at most

$$2^{64}\ell_0KHr.$$

Since $\varepsilon \leq 1/D$, $r \leq 2/(D\varepsilon)$, and $H \leq D^2$. Including the final ℓ calls gives at most

$$\frac{2^{65}\ell_0KD}{\varepsilon} + \frac{12KD}{\varepsilon} + 1 \leq \frac{2^{66}\ell_0KD}{\varepsilon}$$

complex calls. Each is simulated by $q + 1$ real calls. Using $\ell_0 \leq 17q < 32q$ gives the bound $2^{71}q(q + 1)KD/\varepsilon$. The large-error branch is cheaper than this same envelope. Taking the better of this algorithm and exact coordinate-basis reconstruction proves (3).

All choices use a bounded finite number of phase entries, hence finitely many fair bits. D is computable by integer comparisons of D^2 with the rational number H ; no square-root operation on unknown data is needed. Exact rank tests handle singular samples, as specified in Lemma 5.3. $\square$

7 Why a plain global Nyström residual is not enough

The new hierarchy is not just a change of sample count in a standard global PSD Nyström control. The following variance obstruction illustrates the distinction.

Proposition 7.1 (A restricted-family variance floor). *Let $M = uu^T \otimes I_n$ with $\|u\| = 1$. If $0 \preceq B \preceq M$ and $\text{rank } B \leq k < n$, then for independent radius- $\sqrt{n}$ real spherical local probes,*

$$\text{Var}(x^T(M - B)x) \geq \left(\frac{3n}{n+2} - 1\right)(n - k)^2.$$

This is a variance statement about PSD low-rank controls followed by this residual estimator, not a bounded-probability minimax lower bound for RA-11.

Proof. The PSD order forces $\text{range } B \subseteq \text{range } M = u \otimes \mathbb{R}^n$, so $B = uu^T \otimes C$ with $0 \preceq C \preceq I$ and $\text{rank } C \leq k$. Put $D_0 = I - C$, $d_0 = \text{tr } D_0 \geq n - k$. The residual observation factors into independent nonnegative variables $X = (u^T x_1)^2$ and $Y = x_2^T D_0 x_2$. Their means are 1, d_0 , and $\mathbb{E}X^2 = 3n/(n+2)$. Since $\mathbb{E}Y^2 \geq d_0^2$, $\text{Var}(XY) \geq (3n/(n+2) - 1)d_0^2$. $\square$

The hierarchy instead permits indefinite Hermitian controls, preserves the PSD mean of every noisy partial oracle, and charges each residual interaction to a specific approximation error. Those are different mechanisms from global PSD Nyström deflation.

8 Audit, experiments, and reproducibility

8.1 Scope checks

The mathematical construction uses no factorization promise on M . All partial traces $\mathcal{T}_S M$ are PSD because M is PSD, but the residual $M - B$ need not be PSD and is never treated as such. Conditional independence is used only after freezing the stage control, with fresh inside and outside randomness. The metric-median step uses observable candidate distances, not the unknown trace. The original oracle is called only on real rank-one tensor vectors, including when noisy basis sweeps or regression sketches are performed.

The result counts calls, not arithmetic operations, bit complexity, or memory. The algorithm may store a dense $N \times N$ control matrix and may perform an exhaustive column-subset search. It is consequently a query-complexity result, not a claim of an efficient practical solver. The explicit constants are much larger than useful numerical parameters.

8.2 What the bundled checks establish

The exact suite uses Gaussian-rational arithmetic for the partial-trace identities, interpolation, projection updates, and column-subset determinant identities. Exhaustive phase enumerations use bounded integer real/imaginary pairs, with explicit overflow checks. They verify finite instances of isotropy, properness, fourth moments, matrix contraction noise, the component assignment identity, and the final variance budget. Orders through $q = 6$ are checked for interpolation; the exhaustive interaction-noise checks use $n = 2, q = 3$. The schedule inequalities are checked with rational/integer arithmetic for $q = 2, \dots, 64$, $n \in \{2, 3, 10\}$ and representative accuracies. These computations supplement, and do not replace, the all-parameter proofs.

The numerical suite checks the real-only oracle interface, interpolation error, noiseless low-rank regression, zero inputs, basis fallback, and metric-median selection. It also runs 96 reduced-constant trials across dense PSD, rank-two, projector-identity product, and diagonal PSD examples at $(n, q) = (4, 2), (2, 3)$. The full parameters and outputs are recorded in the JSON results. These small experiments can use more calls than exact basis reconstruction. They are diagnostics of implementation and identities, not executions of the enormous certified schedule and not evidence of a universal success rate.

The separate Gaussian-rational reference implementation uses exact elimination, Gram inverses, and exhaustive column selection, without a numerical SVD. Its tests check an exact four-call basis run, preflight query-limit rejection before any calls, and exact low-rank regression identities. The regression tests deliberately use a smaller embedding constant; they are not tests of the universal probability bound. The full certified hierarchy is implemented but has not been executed at its enormous sufficient sample counts.

8.3 Reproduction

From the package root, run

```
bash run_checks.sh
bash build.sh
```

The first command runs all three test suites and regenerates their JSON/log evidence. The build

script compiles this manuscript when a LaTeX installation is available. The source uses NumPy and SymPy; environment versions are recorded separately. The numerical implementation uses floating-point SVD and pseudoinverses for convenience. The exact reference uses Gaussian-rational arithmetic, and the theorem itself permits arbitrary real PSD inputs, not only rational ones.

8.4 Final mathematical status

The all-PSD $O_q(1/\varepsilon)$ upper bound and fixed-order characterization are the new conclusions. The previous product-matrix result and lower bound remain useful but are not promoted beyond their scope. No exhaustive novelty or priority search is claimed. No independent reviewer or formal proof assistant has certified these arguments. Most importantly, the q -dependence remains unmatched; the package is not labeled a full solution of RA-11.

A A self-contained lower bound for the fixed-order corollary

For completeness, we include the ordinary full-vector lower bound needed by Corollary 1.2. It is a specialization of the probability argument in the first package [2], not a new lower-bound claim.

Lemma A.1 (Adaptive Wishart trace posterior). *Let $W = G^T G$, where G is a $d \times d$ matrix of independent standard real Gaussians. After $t < d$ adaptive ordinary matrix-vector queries spanning t directions, conditional on the transcript and the algorithm's random seed,*

$$\text{tr } W = a + X, \quad X \sim \chi_{(d-t)^2}^2, \quad 0 \leq a \leq \text{tr } W,$$

where a is observable and the conditional law of X does not depend on the transcript. Dependent queries may be padded with independent directions for the purpose of this lower bound; this only strengthens the information available to the estimator.

Proof. For a fixed queried coordinate subspace write $G = [G_1 \ G_2]$, with G_1 having t columns. The observations reveal $K = G_1^T G_1$ and $C = G_1^T G_2$. Gaussian orthogonal projections give

$$W = \begin{bmatrix} K & C \\ C^T & C^T K^{-1} C + F^T F \end{bmatrix},$$

where F is a fresh square Gaussian matrix of order $d - t$, independent of the observations. Its squared Frobenius norm is $\chi_{(d-t)^2}^2$, and $a = \text{tr } K + \text{tr}(C^T K^{-1} C) \geq 0$.

For adaptive queries, use induction on the transcript. After s independent directions, the unknown Schur complement is a fresh square Wishart matrix in the unqueried complement. Conditional on the current transcript, the next residual direction is fixed. Orthogonal invariance rotates it to a first coordinate, and the same one-column calculation exposes its column while leaving a fresh square Wishart Schur complement of order $d - s - 1$. This proves the invariant without conditioning on an adaptive subspace as though it were independent of W . The observable term can also be written $\text{tr}(WU(U^T W U)^{-1} U^T W)$ for a basis U of the final queried span; the residual Schur complement is PSD, giving $a \leq \text{tr } W$. $\square$

Lemma A.2 (Shifted log-chi-square anti-concentration). *For $X \sim \chi_\nu^2$, $\nu \geq 2$, $a \geq 0$, the random variable $Y = \log(a + X)$ has a log-concave density and*

$$\text{Var } Y \geq \frac{2\nu}{(a + \nu + 2)^2}.$$

A log-concave real density of variance σ^2 has supremum at most $8/\sigma$.

Proof. The log-density of Y , up to a constant, is $y + (\nu/2 - 1) \log(e^y - a) - (e^y - a)/2$. Its second derivative is

$$-\left(\frac{\nu}{2} - 1\right) \frac{ae^y}{(e^y - a)^2} - \frac{e^y}{2} \leq 0.$$

Gamma integration by parts gives $\text{Cov}(X, \log(a + X)) = 2\mathbb{E}[X/(a + X)]$. Cauchy–Schwarz gives $\mathbb{E}[X/(a + X)] \geq \nu/(a + \nu + 2)$; applying it again to the covariance and using $\text{Var } X = 2\nu$ proves the variance bound.

For the density assertion, let its maximum be J at a mode m , and let the two sides of the superlevel interval $f \geq J/e$ have lengths a_0, b_0 . Its mass implies $a_0 + b_0 \leq e/J$. Log-concavity gives exponential upper tails Je^{-t/a_0} and Je^{-t/b_0} beyond these respective lengths. Consequently

$$\text{Var } Y \leq \int (y - m)^2 f(y) dy \leq J(a_0^3 + b_0^3)(1/3 + 5/e) < 64/J^2.$$

Endpoint modes and truncated sides follow by limits. Thus $J \leq 8/\sigma$. $\square$

Proposition A.3 (Ordinary-query lower bound). *Relative PSD trace estimation with success at least $2/3$ in dimension N requires at least $\min\{N, 1/\varepsilon\}/80000$ ordinary matrix-vector queries.*

Proof. Put $d = \min\{N, \lfloor 1/(10000\varepsilon) \rfloor\}$ and suppose first that $d \geq 4$. Embed the Wishart prior in a known d -dimensional coordinate subspace. Suppose at most $t \leq d/4$ queries are used. After padding, apply Lemma A.1. Since $\mathbb{E}a \leq d^2$, the event $a \leq 10d^2$ has probability at least $9/10$. On this event, $\nu = (d - t)^2$ and Lemma A.2 imply

$$\text{Var}(\log \text{tr } W \mid \text{transcript}) \geq \frac{1}{500d^2}.$$

Its conditional density is at most $8\sqrt{500}d$. For any positive transcript-based estimate, relative-error success confines $\log \text{tr } W$ to an interval of length $\log((1 + \varepsilon)/(1 - \varepsilon)) \leq 3\varepsilon$. Nonpositive estimates never succeed under this positive prior. The total success probability is therefore at most

$$1/10 + 24\sqrt{500}\varepsilon d \leq 1/10 + 24\sqrt{500}/10000 < 2/3.$$

The prior is independent of the algorithm’s random seed, so the bound holds after averaging over that seed. Uniform success on every fixed PSD matrix would imply average success at least $2/3$, a contradiction. Thus $t > d/4$ is necessary. For $d \geq 4$, $d \geq \min(N, 1/\varepsilon)/20000$, giving the claimed constant. If $d < 4$, then $\min(N, 1/\varepsilon) < 40000$ in the present setting $N \geq 4$. At least one call is necessary: a zero-call output cannot fall with probability $2/3$ in both the zero-trace event and the disjoint success interval of a positive trace input. This covers the remaining case. $\square$

B Reference list and provenance

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